

URL, DNS, HTTP and HTTPS — Detailed Lecture Notes

JKSSB Computer Awareness

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35.1 What a URL is

A URL (Uniform Resource Locator) is the full address of one resource on the Internet: one page, file, image, or service. Typing a URL tells the browser both how to connect and exactly what to fetch.

Definition (URL). *The complete address of a single Internet resource, naming the protocol (scheme), the host, and the location of the resource on that host, including the optional port, path, and query.*

Term	Meaning in this lesson
URL	The full address of one resource, such as a page or file.
Scheme	The protocol word before the colon, such as https or http.
Domain name	The readable host name, such as www.example.in.
TLD	Top-Level Domain: the last label, such as .in or .com.
Subdomain	A prefix section of the name, such as www or mail.
Port	The door number on the server (80 for HTTP, 443 for HTTPS).
Path	The location of the page inside the site.
Query	Parameters after a ?, such as ?topic=dns.
IP address	The numeric address of a device (IPv4 is 32-bit, IPv6 is 128-bit).
DNS	Domain Name System: translates domain names into IP addresses.
HTTP	Hypertext Transfer Protocol: carries web pages.
HTTPS	Hypertext Transfer Protocol Secure: HTTP protected by TLS.
TLS	Transport Layer Security: encrypts the exchange (successor of SSL).

The rest of this lesson uses these terms in one consistent sense. A **URL** is the whole address; the **domain name** is only the host part inside it. **DNS** translates that host name into an **IP address**. **HTTP** moves the page, and **HTTPS** is HTTP carried inside a **TLS**-encrypted channel.

35.2 Anatomy of a URL

Consider the example used throughout this lesson:

`https://www.example.in:443/study/computer?topic=dns&lang=en`

Read it left to right. The **scheme** is `https`: the protocol the browser must speak. The `://` is only a separator between the scheme and the host. The host is `www.example.in`, in which `www` is the **subdomain**, `example` is the registered domain, and `.in` is the **TLD** (Top-Level Domain). Common TLDs include `.com`, `.org`, `.edu`, `.gov`, and country labels such as `.in`.

After the host comes the optional **port**, `:443`: the door number on the server. The **path**, `/study/computer`, locates the page inside the site. The **query**, `?topic=dns&lang=en`, passes extra parameters to the server. Port and query are optional: when the port is missing, the browser fills in the default for the scheme.

Definition (TLD (Top-Level Domain)). *The last label of a domain name, such as `.com`, `.org`, `.edu`, `.gov`, or a country label such as `.in`.*

Example 35.2.1. *Split `mail.example.com`. The subdomain is `mail`, the registered domain is `example`, and the TLD is `.com`. One registered domain can carry many such subdomains (`www`, `mail`, `blog`), each pointing to a different service, while the domain itself stays the same.*

35.3 Ports, paths, and queries

The **port** selects which service on the server receives the connection. Port **80** is the default for HTTP and port **443** is the default for HTTPS, which is why both numbers are usually hidden in the address bar. The **path** names the page or file within the site, and the **query** carries name–value pairs after a `?` mark. Neither the port nor the query changes which host is contacted; the host is fixed by the domain name alone.

35.4 IP addresses: the real destination

An IP (Internet Protocol) address is the numeric address every device uses to send and receive data on the network. People type names; machines route by numbers.

Definition (IP address). *The numeric address of a device on a TCP/IP network, used for routing data to that device.*

IPv4 (Internet Protocol version 4) is **32-bit** and is written as four decimal groups, for example `192.0.2.44`. **IPv6** (Internet Protocol version 6) is **128-bit** and is written in hexadecimal groups. It is never 256-bit; that figure is a repeated wrong option. IPv6 exists because the IPv4 space is too small for the number of connected devices.

35.5 DNS: the phone book of the Internet

DNS (Domain Name System) is the distributed naming system that translates domain names into IP addresses. It does not carry page content and does not assign addresses to devices; it only answers the question “which IP address belongs to this name?”

Definition (DNS). *The hierarchical, distributed system that maps each domain name to the IP address of its server, with answers cached to speed up repeat visits.*

Resolution runs as a short chain. The browser first checks its own cache for the name. If the name is unknown, the **resolver** (usually run by the ISP — Internet Service Provider) asks the DNS hierarchy: the root servers, then the TLD servers for labels such as `.in`, then the domain’s own authoritative server. That authoritative server returns the IP address, the resolver caches the answer for a fixed time, and the browser receives the number it can connect to.

Example 35.5.1. *On the office desktop, typing `www.example.in` first triggers a DNS lookup. The resolver walks the hierarchy, learns the address (say `192.0.2.44`), caches it, and only then does the browser open a connection to that number. Typing the IP address directly would skip DNS, but nobody memorises numbers for every site.*

35.6 HTTP: carrying web pages

HTTP (Hypertext Transfer Protocol) is the request–response protocol of the web. The browser sends a request naming the path (and query); the server returns the page. HTTP runs by default over port 80. Plain HTTP provides no encryption: the request and the page travel in readable form, so anyone on the path between browser and server can read them.

Definition (HTTP). *The protocol by which a web client requests a resource and the server returns it. Plain HTTP does not encrypt the exchange.*

35.7 HTTPS and TLS: the secure form

HTTPS (Hypertext Transfer Protocol Secure) is HTTP protected by TLS. TLS (Transport Layer Security) is the successor of the older SSL (Secure Sockets Layer); the phrase “SSL/TLS” survives only as a name for the same family of protection. Before any page moves, the browser and server run a TLS handshake: the server proves its identity with a certificate, and the two sides agree on encryption keys. The HTTP request and response then travel inside that encrypted channel.

Definition (HTTPS). *HTTP carried inside a TLS-encrypted channel, by default over port 443 and shown with a padlock in the address bar.*

TLS provides three things at once: **confidentiality** through encryption (only the two ends can read the data), **integrity** (tampering is detected), and server **authentication** (the certificate ties the site to its claimed name). HTTPS therefore protects the trip; it is not a different page language.

35.8 The full flow: from typed URL to secure page

Put the pieces together for `https://www.example.in/study/computer`. The browser parses the URL: scheme `https`, host `www.example.in`, path `/study/computer`. DNS resolves the host to an IP address. The browser opens a connection to that IP on port 443, the HTTPS default. The TLS handshake secures the channel, and the HTTP request for the path travels inside it. The server returns the page, and the browser renders it with the padlock shown.

35.9 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. A **URL** addresses one resource; the **domain name** is only the host part inside it.
2. **DNS** translates names into IP addresses. It does not assign addresses and does not carry page content.
3. **IPv4** is 32-bit; **IPv6** is 128-bit, never 256-bit.
4. **HTTP** uses port 80; **HTTPS** uses port 443. Do not reverse them.

5. **HTTPS** is HTTP over **TLS**. HTTP alone gives no encryption.
6. The **scheme** is the protocol word (**https**), not the domain name that follows it.

Example 35.9.1. *An item asks which part of `https://mail.example.com/inbox` is the subdomain. Trace the host `mail.example.com` from the right: the TLD is `.com`, the domain is `example`, and the remaining prefix `mail` is the subdomain. If the stem instead asks for the scheme, the answer is `https`; if it asks for the default port of that scheme, the answer is `443`.*

Exercises

1. Expand URL, DNS, HTTP, HTTPS, TLS, IP, TLD, and ISP.
2. Label every part of `https://blog.example.org:443/notes/net?ch=1`.
3. Distinguish a URL from a domain name.
4. Describe the steps of DNS resolution when a name is not cached.
5. Contrast HTTP with HTTPS on security, default port, and address-bar form.

Solutions

1. URL — Uniform Resource Locator; DNS — Domain Name System; HTTP — Hypertext Transfer Protocol; HTTPS — Hypertext Transfer Protocol Secure; TLS — Transport Layer Security; IP — Internet Protocol; TLD — Top-Level Domain; ISP — Internet Service Provider.
2. Scheme `https`; subdomain `blog`; domain `example`; TLD `.org`; port `443`; path `/notes/net`; query `?ch=1`.
3. A URL is the full address of one resource (scheme, host, port, path, query). A domain name is only the readable host part inside it.
4. The browser checks its cache; then the resolver asks the DNS hierarchy (root, then TLD, then the authoritative server), which returns the IP address; the resolver caches it and hands it to the browser.
5. HTTP carries pages without encryption over port 80 (`http://`); HTTPS is HTTP protected by TLS over port 443 (`https://` with a padlock).